

GUTA 1 (parte 1)

1.1) $\Omega = \{1, 2, 3, 4, 5, 6\} \rightarrow$ espacio muestral

a) menor $\mathcal{A}$ tal que $\{1, 2, 3\} \in \mathcal{A}$

La menor algebra seria $\mathcal{A} = \{\Omega, \emptyset\}$, pero tengo que agregar el $\{1, 2, 3\}$ y su complemento, y la UNIÓN!

$$\mathcal{A}_1 = \{\emptyset, \Omega, \{1, 2, 3\}, \{4, 5, 6\}\}$$

$$(\emptyset \cup \Omega) = \Omega$$

$$(\{4, 5, 6\} \cup \Omega) = \Omega$$

$$(\{1, 2, 3\} \cup \{4, 5, 6\}) = \Omega$$

$$(\{1, 2, 3\} \cup \Omega) = \Omega$$

b) $\{1, 2\}, \{3, 4\}, \{5, 6\}$

$$\mathcal{A}_2 = \{\emptyset, \Omega, \{1, 2\}, \{3, 4\}, \{5, 6\}, \{3, 4, 5, 6\}, \{1, 2, 5, 6\}, \{1, 2, 3, 4\}\}$$

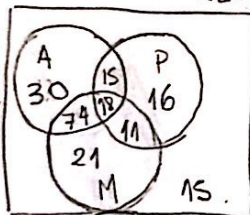
$\{1, 2\} \cup \{3, 4\} = \{1, 2, 3, 4\} \rightarrow$ CADA UNIÓN ya está expresada en el complemento

1.2) 200 ESTUDIANTES

UN ESTUDIANTE al azar

↓
ESPACIO EQUIPROBABLE

Laplace
 $P = \frac{\text{FAVOR}}{\text{TOTALES}}$



- 137 cursan Algebra II
 - 60 PROBA
 - 124 MATERIALES
 - 33 algebra y proba
 - 29 proba y materiales
 - 92 materiales y algebra
 - 18 (así)
- $\Omega = 200$

a) curse algebra II ^{UNIÓN} o proba

b) no curse A ni P $\rightarrow$ No ALGEBRA y No PROBA

c) curse alguna de los 3

d) curse solo 1/3

e) no curse ninguno.

$$\Omega = \{200 \text{ est}\} \quad \# \Omega = 200$$

A = curse Alg M = curse Mat
P = curse Prob.

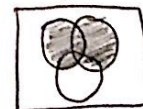
$$\# A = 137$$

$$\# P = 60$$

$$\# M = 124$$

a) $P(A \cup P) = P(A) + P(P) - P(A \cap P)$ por los conj no son disjuntos.

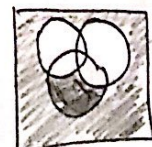
$$P(A \cup P) = \frac{137}{200} + \frac{60}{200} - \frac{33}{200} = \frac{41}{50} = 0.82$$



b) $P(\overline{A \cap P}) = P(\overline{A \cup P}) = 1 - P(A \cup P) = 1 - (P(A) + P(P) - P(A \cap P)) = 1 - 0.82$

↑
ARICO MORGAN

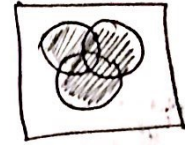
$$P(\overline{A \cap P}) = 0.18$$



$$P(A \cup P \cup M) = P(A) + P(P) + P(M) - P(A \cap P) - P(A \cap M) - P(P \cap M) + P(A \cap P \cap M) =$$

$$= \frac{137}{200} + \frac{60}{200} + \frac{124}{200} - \frac{33}{200} - \frac{92}{200} - \frac{29}{200} + \frac{18}{200} =$$

$$P(A \cup P \cup M) = \frac{185}{200} = \frac{37}{40} = 0.925$$



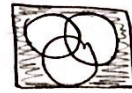
d) Solo. cuase una de las 3. $\rightarrow$ algebra o mat o proba

$$\left. \begin{aligned} P(\text{solo algebra}) &= \frac{30}{200} \rightarrow A \cap \bar{P} \cap \bar{M} \\ P(\text{solo } P) &= \frac{16}{200} \rightarrow P \cap \bar{A} \cap \bar{M} \\ P(\text{solo mat}) &= \frac{24}{200} \rightarrow M \cap \bar{A} \cap \bar{P} \end{aligned} \right\} \text{la unión de esto.}$$



$$P\left[(A \cap \bar{P} \cap \bar{M}) \cup (P \cap \bar{A} \cap \bar{M}) \cup (M \cap \bar{A} \cap \bar{P})\right] = \frac{67}{200} = 0.335$$

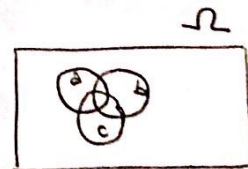
e) $P(\bar{A} \cap \bar{P} \cap \bar{M}) = \frac{15}{200} = 0.075$



Ni A, Ni P, Ni MAT.

1.3) $\Omega = \{a, b, c\}$, $A \subset \Omega$, $P(A) = \sum_{w \in A} P(w)$ con $P(a) = \frac{1}{2}$, $P(b) = \frac{1}{3}$, $P(c) = \frac{1}{6}$
 $\rightarrow$ A está incluido en Ω
 8 subconjuntos de Ω .

$A = \{a, b, c\}$ $D = \{b\}$ $G = \{b, c\}$
 $B = \{\emptyset\}$ $E = \{c\}$ $H = \{a, c\}$
 $C = \{a\}$ $F = \{a, b\}$



$$P(A) = \sum_{w \in A} P(w) = P(a) + P(b) + P(c) = 1$$

$\downarrow$
elementos de A

$$P(B) = P(\emptyset) = 0$$

$$P(G) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

$$P(C) = P(a) = \frac{1}{2}$$

$$P(F) = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

$$P(D) = P(b) = \frac{1}{3}$$

$$P(H) = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$$

$$P(E) = P(c) = \frac{1}{6}$$

DADO EQUIVADO $\rightarrow$ espacio equiprobable

$\hookrightarrow$ ARROJA 2 VECES $\Omega: \{(m, m) / m, m \in \{1, 2, 3, 4, 5, 6\}\}$

$$\Omega: \{1, 2, 3, 4, 5, 6\}^2$$

$$\#\Omega = 36$$

$$q = 2^2 \text{ conj. posibles}$$

a) SUMA SEA 7

1-6

2-5

3-4

4-3

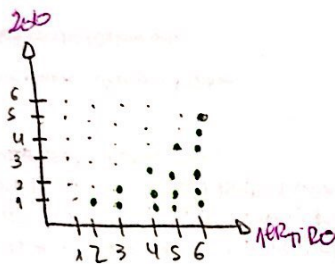
5-2

6-1

$$P(\text{SUMA SEA } 7) = 6/36 = 0,167$$

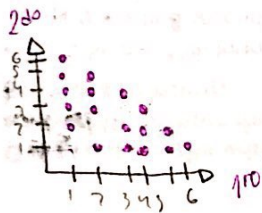
$$P = \frac{\text{FAV}}{\text{POSIBLES}}$$

b)



$$P(\text{suma} > 7) = 9/36 = 0,25$$

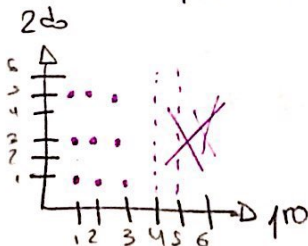
c) 2 RESULT $\neq$ Y NO SUMA $> 7 = C$



1-2
1-3 x2
1-4
1-5
1-6
2-3
2-4 x2
2-5
3-4 x2

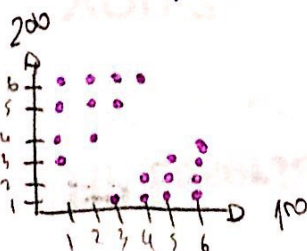
$$P(C) = 18/36 = 0,5$$

d) 1er < 4 , 2do IMPAR = D

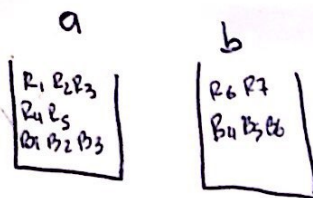


$$P(D) = 9/36 = 0,25$$

e) $|1ro - 2do| > 1 = E$



$$P(E) = 20/36 = 0,56$$



AZAR

$$\Omega = \{(m, m) / (m, m) \in \{a, b\}\}$$

2) A = 2 rojas

R₁ R₆ R₂ R₆ R₃ R₆ R₄ R₆ R₅ R₆
R₁ R₇ R₂ R₇ R₃ R₇ R₄ R₇ R₅ R₇

$$P(A) = \frac{10}{40} = 1/4$$

$$\{12345678\} \times \{1, 2, 3, 4, 5\} =$$

$$\# \Omega = 40$$

b) B = mismo color

10 de colores rojos

$$P(B) = \frac{19}{40} = 0.475$$

B₁ <

B₂ < + 9

B₃ <

c) R₁ < R₂ < R₃ < R₄ < R₅ 3 x 5 = 15

B₁ < B₂ < B₃ < 3 x 2 = 6 15 + 6 =

$$P(C) = \frac{21}{40} = 0.525$$

OTRA FORMA

eventos indep.

d) 8 < x 3 = 24

$$P(D) = \frac{24}{40} = \frac{3}{5} = 0.6$$

1.6) 4 dados → NINGÚN 1

$$\Omega = \{(a, b, c, d) : a, b, c, d \in \{1, 2, 3, 4, 5, 6\}\}$$

$$\# \Omega = 6^4 = 1296$$

A: No sale 1

$$P(A) = 1 - P(\bar{A}) = 1 - (5/6)^4 \approx 0.52 \text{ sale 1}$$

Hay otra más chances que salga uno en los 4 dados.

0.48 no sale 1

(4)

a) $\Omega = \{X_1, X_2, X_3, \dots, X_n\}$

b) $P(A_1), P(A_2), P(A_{2016}) \rightarrow A_m$, en m sale el primer 6.

$P(A_1) \rightarrow$ en el 1^{er} tirado sale el 6.

$$P(A_1) = 1/6$$

$$P(A_2) = \frac{5^1 \cdot 1}{6^2} = 0,047$$

$$P(A_{2016}) = \frac{5^{2015} \cdot 1}{6^{2016}} \approx 0$$

c) $B_m \rightarrow N > m \rightarrow$ no sale 6 en m tirados.

$$P(B_1) = 5/6$$

$$P(B_8) = \frac{5^8}{6^8} = 0,23$$

$P(B_{2016}) \approx 0$. Hay pocas prob que tirando

2016 veces ninguna sale 6.

$\rightarrow$ m es muy grande.

d)

$$B_{m+1} \subset B_m$$

B_{m+1} está incluido en B_m

$$P(B) = \lim_{m \rightarrow \infty} P(B_m) = \lim_{m \rightarrow \infty} \left(\frac{5}{6}\right)^m = 0$$

$$P(B_{m+1}) = \left(\frac{5}{6}\right)^{m+1}$$

$$P(B_m) = \left(\frac{5}{6}\right)^m$$

Si $P(B_{m+1}) < P(B_m)$

entonces.

$$= \left(\frac{5}{6}\right)^m \cdot \frac{5}{6} = A \cdot \left(\frac{5}{6}\right)^m = \left(\frac{5}{6}\right)^m = A \cdot 1$$

si $m = 3$

$$P(B_4) = 0,148 < P(B_3) = 0,58$$

$$\boxed{P(B_{m+1}) \subset B_m}$$

e) $B = \bigcap_{m=1}^{\infty} B_m$

AY. CONTINUADO

$$\rightarrow \lim_{m \rightarrow \infty} P(B_m) = \lim_{m \rightarrow \infty} \left(\frac{5}{6}\right)^m = 0$$



nunca sale un 6.



Probabilidad del vacío es cero.

$\rightarrow$ puede ocurrir, pero no va a pasar.

⑤

$$\Omega = \{0, 1, 2, 3, \dots\} ; A \subset \Omega$$

$$P(A) = \sum_{m \in A} P(m) = \sum_{m \in A} \frac{c}{m!} \quad \text{Hallar } c \text{ y calcular } P(\{0, 2, 4, 6, \dots\})$$

$$\text{Yose que } P(\Omega) = 1$$

$$\sum_{m=0}^{\infty} \frac{c}{m!} = 1$$

$$c \cdot \sum_{m=0}^{\infty} \frac{1}{m!} = 1$$

$$c \cdot e^1 = 1$$

$$\boxed{c = 1/e}$$

$$m = 2k$$

$$\sum_{m=0}^{\infty} \frac{c}{m!} = \sum_{k=0}^{\infty} \frac{e^{-1}}{(2k)!} = e^{-1} \cdot \cosh(1) = e^{-1} \left(\frac{e^1 + e^{-1}}{2} \right)$$

coseno
hiperbólico

1.9)

$$P = \frac{1}{3 \cdot 2^{k-1}} \quad \text{con } k = \text{cantidad de personas}$$

Juan vs Pedro
Gaudius vs Maria

a)

$$\Omega = \{JJ, PP, JMM, \dots\} \quad \text{resultados posibles}$$

A: gana Juan

$$\left. \begin{array}{l} JJ \quad PMJJ \\ JMPJJ \quad (PMJ)^k J \\ (JMP)^k JJ \end{array} \right\}$$

$$A = \bigcup_{k=0}^{\infty} \{(JMP)^k JJ\} \cup \bigcup_{k=1}^{\infty} (PMJ)^k J$$

todos los
posibilidades
gana Juan

B: gana Pedro

$$\left. \begin{array}{l} PP \quad JMPP \\ (PMJ)^k PP \quad (JMP)^k P \end{array} \right\}$$

$$B = \bigcup_{k=0}^{\infty} ((PMJ)^k PP) \cup \bigcup_{k=1}^{\infty} ((JMP)^k P)$$

C: gana Maria

$$\left. \begin{array}{l} JMM \quad PMM \\ (JMP)^k JMM \quad (PMJ)^k PMM \end{array} \right\}$$

$$C = \bigcup_{k=0}^{\infty} ((JMP)^k JMM) \cup \bigcup_{k=0}^{\infty} ((PMJ)^k PMM)$$

$$\Omega = \{A \cup B \cup C\}$$

6

composicion disjuntos $\downarrow$ paridos $\downarrow$ paridos $\downarrow$ por Gauss

$$P(A) = P\left(\bigcup_{k=0}^{\infty} (JMP)^k\right) + P\left(\bigcup_{k=1}^{\infty} (PM)^k\right) =$$

$$= \sum_{k=0}^{\infty} \frac{1}{3 \cdot 2^{(3k+2)-1}} + \sum_{k=1}^{\infty} \frac{1}{3 \cdot 2^{(3k+1)-1}} =$$

$$= \frac{1}{3} \sum_{k=0}^{\infty} \left(\frac{1}{8}\right)^k + \frac{1}{3} \sum_{k=1}^{\infty} \left(\frac{1}{8}\right)^k \leftarrow \text{serie geometrica}$$

$$= \frac{1}{6} \left(\frac{1}{1-\frac{1}{8}}\right) + \frac{1}{3} \left(\frac{\frac{1}{8}}{1-\frac{1}{8}}\right) = \frac{5}{24}$$

$$[P(B) = 5/24 \text{ (es igual a forma)}]$$

$$P(C) = \sum_{k=0}^{\infty} \frac{1}{3 \cdot 2^{(3k+3)-1}} + \sum_{k=0}^{\infty} \frac{1}{3 \cdot 2^{(3k+3)-1}}$$

$(3k+3)$

$$P(C) = \frac{1}{3} \cdot \left(\frac{1}{2}\right)^2 \sum_{k=0}^{\infty} \left(\frac{1}{8}\right)^k + \frac{1}{12} \sum_{k=0}^{\infty} \left(\frac{1}{8}\right)^k$$

$$P(C) = \frac{1}{12} \cdot \left(\frac{1}{1-\frac{1}{8}}\right) + \frac{1}{12} \left(\frac{1}{1-\frac{1}{8}}\right)$$

$$[P(C) = 4/21]$$

c)

$$(JMP)^k$$

$$P(\text{Juan gana} \cap \text{no hay ganador}) = 1 - P(\text{Juan no gana} \cup \text{hay ganador})$$

$$1 - 5/9 \quad \quad \quad 4/21 + 10/21 = 14/21$$

Hay intersección.

$$P(\sim) = 1 - \left(\frac{10}{9}\right) + P(\text{Juan no gana} \cap \text{hay ganador})$$

$\Rightarrow$

(7)

$$P(\sim) = 1/9 + P(\text{no hay} \cap \text{hay}) = 2/9$$

$$\begin{aligned} A: (PMJ)^K J &\rightarrow \frac{1}{3} \cdot \left(\frac{1/8}{1-1/8}\right) = 7/24 \\ B: (PMJ)^K PP &\rightarrow \frac{1}{3} \cdot \frac{1}{2} \left(\frac{1}{1-1/8}\right) = 4/21 \\ C: (PMJ)^K PMM &\rightarrow \frac{1}{3} \left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{1-1/8}\right) = 2/21 \end{aligned} \quad \left. \vphantom{\begin{aligned} A: \\ B: \\ C: \end{aligned}} \right\} 0,33.$$

$$d) 1 - 14/21 = 7/21$$

$$\boxed{P(\text{ninguno siempre}) = 7/21}$$

$$1.10) [0,1] \rightarrow P[0,1] = b-a \text{ longitud del intervalo.}$$

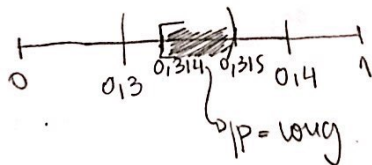
$$a) 0,314$$

$$\Omega = [0,1]$$

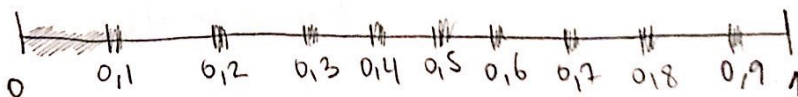
$$A = 2^{[0,1]}$$

$$P(0,314) = P(a_1=3 \wedge a_2=1 \wedge a_3=4)$$

$$\downarrow \text{en que intervalo está?} = 0,315 - 0,314 = 10^{-3}$$



b)



A: el cero no aparece en los 10's 4. $\bar{A}$ = si aparece.

$$P(A) = 1 - P(\bar{A}) = 1 - (9^3 \cdot 10^{-4}) = 0,9271$$

$$\begin{aligned} c) P(A) &= 1 - P(\bar{A}) = 1 - P\left(\bigcup_{m=1}^{\infty} A_m\right) = 1 - \sum_{m=1}^{\infty} P(A_m) = 1 - \sum_{m=1}^{\infty} 9^{m-1} \cdot 10^{-m} = \\ &= 1 - 9 \cdot \sum_{m=1}^{\infty} \left(\frac{9}{10}\right)^m = 1 - \frac{1}{9} \cdot \left(\frac{9/10}{1-9/10}\right) = 0 \end{aligned}$$



8	9
4	5 6 7
0	1 2 3

AZAR CON REPOSICIÓN 5

a) 10 10 10 10 10

$$\Omega = \{(a_1, b_1, c_1, d_1, e_1) : \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}\}$$

$$\# \Omega = 10^5$$

1. A = Las 5 sean iguales

Regla
del producto

$$\begin{array}{c} \frac{10}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \\ \text{Tenemos la opción de elegir cualquiera.} \end{array}$$

Solo puedo elegir la que tenga.

$$\#A = 10 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 10$$

$$P(A) = \frac{\#A}{\# \Omega} = \frac{10}{10^5}$$

$$P(A) = \frac{1}{10^4}$$

2. 1, 3, 5, 7, 9.

$$\frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \left[P(B) = \frac{1}{10^5} \right]$$

3. 5 4 3 2 1 $\#C = 5! = 120$

$$P(C) = \frac{120}{10^5} = 0,0012$$

4. $\#D = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30240$

$$P(D) = 0,3024$$

b) SIN REPO

$$\# \Omega = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6$$

$$\# \Omega = 30240$$

1. evento imposible
 $P(A) = 0$

3. $P(C) = \frac{120}{30240} = \frac{1}{252}$

2. $\frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad \frac{1}{\downarrow} \quad P(B) = \frac{1}{30240}$

4. 10 9 8 7 6 $P(D) = 1$

93aR

CON REPO PE
 $\# \Omega = 29^{12}$

Porque de los 29 que podríamos, solo 12 veces se volvió el número.

$$P(A) = \frac{29 \cdot 28^{11}}{29^{12}} = 0,6797$$

$$b) P(B) = \frac{29!}{29^{12} 17!} = \frac{29 \cdot 28 \cdot 27 \cdot 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21 \cdot 20 \cdot 19 \cdot 18}{29^{12}} = 0.07$$

1.15) 13 piratas y 3 pueros $\begin{matrix} P \\ M \\ G \end{matrix}$ azar

2) 4-OP
4-SM
5-OG

$\Omega = \{ \text{cada piraeta elige los 3 pueros} \} = 3^{13}$

3 3

$$\#A = \frac{13}{4} \cdot \frac{12}{3} \cdot \frac{11}{2} \cdot \frac{10}{1} \cdot \frac{9}{4} \cdot \frac{8}{3} \cdot \frac{7}{2} \cdot \frac{6}{1} \cdot \frac{5}{5} \cdot \frac{4}{4} \cdot \frac{3}{3} \cdot \frac{2}{2} \cdot \frac{1}{1}$$

$$\#A = \frac{13!}{4!4!5!} = 90090$$

$$P(A) = \frac{90090}{315} = 0,0565$$

b) $6 \rightarrow P$ UNION $6 \rightarrow P$
 $6 \rightarrow M$ $7 \rightarrow M$ } union
 $1 \rightarrow G$ disjunta.

$$\#B_n = \frac{13!}{6!6!1!}$$

$$P(B) = P(B_1 \cup B_2) = P(B_1) + P(B_2)$$

$$\#B_2 = \frac{B_1}{63700}$$

$$P(B) = \frac{\#B_1}{\#A} + \frac{\#B_2}{\#A} = 7.53 \times 10^{-3} + 1.107 \times 10^{-3} = 8.637 \times 10^{-3}$$

5 POGOM
4 POGOM
4 POGOM

$$\#\Omega = 3^{13}$$

$$\#C = \binom{13}{5} \binom{8}{4} \binom{4}{4} 3! = 540540$$

↓
tiene 3 separaciones

$$P(C) = \frac{540540}{3^{13}} = 0,134$$

d) ?

1.16) 7 gatos y 5 cajas [EQUIPROBABLE]

↓
indistinguibles → "única forma de ordenarlos"

5 cajas y 7 gatos

11010110101 → una tener 4 ceros
y 7 unos.

$$\#\Omega = \frac{11!}{7!4!}$$

a) 1ra caja → 2 gatos y última cero

110 - 0 - 0 - 0 ^{NADA}
quiero 5 gatos como quiera → lo que quiero

$$\#A = \binom{7}{2} = \frac{(5+2)!}{2!5!}$$

$$P(a) = \frac{\#A}{\#\Omega} = \frac{7!}{5!2!} \cdot \frac{4!7!}{11!}$$

$$P(a) = 0,06$$

2) outra coisa contenga mais de 3 zeros.

a) $_0_0_0_0_0_$

b) $_0_0_0_0_0_$

c) $_0_0_0_0_0_$

d) $_0_0_0_0_0_$

sumo
todo

$$\# \Omega = \frac{11!}{4!7!}$$

a) $\# A \rightarrow \begin{matrix} 3 \text{ 1's} \\ 3 \text{ 0's} \end{matrix} \quad \# A = \frac{6!}{3!3!}$

b) $\# B \rightarrow \begin{matrix} 2 \text{ 1's} \\ 3 \text{ 0's} \end{matrix} \quad \# B = \frac{5!}{3!2!}$

c) $\# C = \frac{4!}{3!1!}$

d) $\# D = \frac{3!}{3!0!} = 1$

$$P(b) = \frac{\frac{6!}{3!3!} + \frac{5!}{3!2!} + \frac{4!}{3!1!} + \frac{3!}{3!0!}}{\frac{11!}{4!7!}}$$

$$P(b) = \frac{20 + 10 + 4 + 1}{330} = \frac{4}{66}$$

$$P(b) = 0,106$$

12

1.77

25 piezas $\rightarrow$ k defectuosos

5 al azar $\rightarrow$ si hay al menos 1 defect se rechaza

a) $k=3 \rightarrow$ hay 3 defectuosos $\rightarrow$ ¿P: pose inspección?

total $\rightarrow$ 25 piezas $\left\{ \begin{array}{l} 3 \text{ defectuosos} \\ 22 \text{ bien} \end{array} \right\}$ elijo 5

de 22 no defectuosos $\leftarrow$ $P = \frac{\binom{22}{5} \binom{3}{0}}{\binom{25}{5}} =$ de 3 defectuosos elijo 0

de 25 $\leftarrow$ $\binom{25}{5}$

22 $\checkmark \rightarrow$ X
3 $\checkmark \rightarrow$ Y

Para que pose mere que haber 0 defectuosos
 $\downarrow$
si hay 1 chuan.

$$P = \frac{22!}{5!17!} \cdot \frac{3!}{3!0!} = \frac{26334}{53130} = 0.496$$

b) $P(k)?$ $P = \frac{\binom{25-k}{5} \binom{k}{0}}{\binom{25}{5}} \rightarrow$ "pose inspección"

c) más de 5 defectuosos $\rightarrow$ acepte una partida con 5^+ defectuosos

1.18

A, T, C, G $\rightarrow$ 4 letras

a) "A precede a T" y "C precede a G" son ind.

que ent
años

A

Para que sean ind

B CG...

$$P(A \cap B) = P(A) P(B)$$

$$P(A) = \frac{12}{24} = \frac{1}{2}$$

prob de que
AT

$$P(B) = \frac{12}{24} = \frac{1}{2}$$

$$\# \Omega = 24 = 6 + 6 + 6 + 6$$

• <u>ATCG</u>	TACG	• CATG
• <u>ATGC</u>	TAGC	• CAGT
• <u>ACGT</u>	TGCA	C TAG
• <u>ACTG</u>	TCAG	C TGA
• <u>AGTC</u>	TGAC	• C GAT
• <u>AGCT</u>	TGCA	C GTA

$$P(A) P(B) = 1/4$$

conclusion

$P(A \cap B) \rightarrow$ A y T a la vez
y G y C a la vez

$$P(A \cap B) = 6/24 = 1/4$$

son indep ✓

GACT
GATC
GTCA
GTAC
G CAT
GC TA

b) "AT" y "CG" no son ind

$$P(C) = 4/24$$

$$P(D) = 4/24$$

$$P(C) P(D) = 1/36$$

$$P(C \cap D) = 2/24 = 1/12 \quad - \quad \neq$$

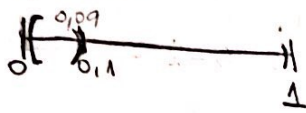
4.19)

$(0,1) \rightarrow \text{azar}$

$P(a,b) = b-a$ ← *long del intervalo*

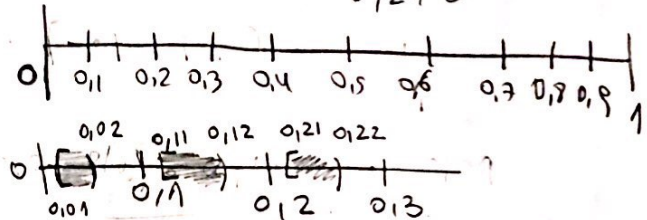
a) "el 1º es cero" y "el 2º es 1" son *ind*

A: 0, 0...



$P(A) = 0,109$

B: 0, 1... 0,010
0,110
0,1210

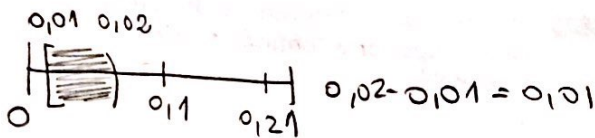


0,01...

$0,02 - 0,01 = 0,01 \rightarrow$ *esto es 9 veces*

$P(C) = P(A) \cap P(B)$

$P(B) = 0,109$



Ⓢ P. 17

Ⓢ

1.20

2 veces $\rightarrow \# \Omega = 36$

a) A: "1er resultado par"

B: "2do resultado par"

C: "suma resultados par"

A, B, C son ind
de los, pero
no general

1 tiro $\rightarrow 6$ resultados $\leftarrow P(A) = 1/2$

↓

3 par

$P(B) = 1/2$

$P(C) = 18/36 = 1/2$

DADO 1

D	1	2	3	4	5	6
A	1	2	3	4	5	6
B	2	3	4	5	6	7
C	3	4	5	6	7	8
D	4	5	6	7	8	9
E	5	6	7	8	9	10
F	6	7	8	9	10	11
G	7	8	9	10	11	12

$$P(A) P(B) P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

ambos par $\leftarrow P(A \cap B \cap C) = 9/36 = 1/4 \neq$
suma par

ambos par $\leftarrow P(A \cap B) = 9/36 = 1/4 = P(A) P(B) = 1/4 \checkmark$
 $P(A \cap C) = 9/36 = 1/4 \checkmark$

no par
suma par $\leftarrow P(B \cap C) = 9/36 = 1/4 \checkmark$

c) A: 1er resultado es 1, 2 o 3

C: suma es 9

B: 1er resultado es 3, 4 o 5

$$P(A \cap B \cap C) = P(A) P(B) P(C) \rightarrow \text{no son ind}$$

1er 3. $\leftarrow P(A \cap B) = 6/36 = 1/6$ $P(A) \cdot P(B) = 1/4$

$\downarrow$ $\downarrow$
 $18/36$ $18/36$
 $\neq$

16

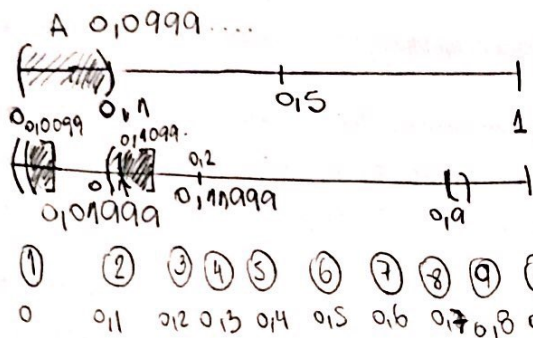
19) $(0,1) \rightarrow 232R$

a) A : "el primer dígito es cero" $\left. \begin{array}{l} \text{0,0...} \\ \text{0,01...} \end{array} \right\}$ son independientes
 B : "el segundo dígito es 1"

$\bar{B}$ ←

$P(\text{intr}) = \text{distancia}$

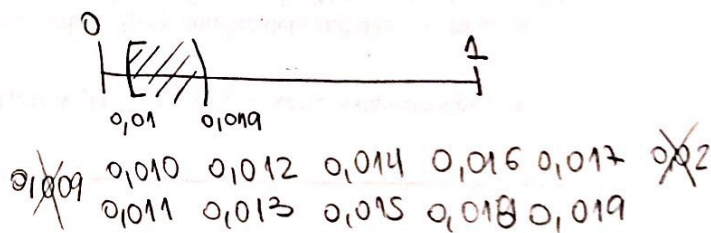
$P(A \cap B) = P(A)P(B)$



$P(A) = P(0, 0.1) = 0.1$

$P(B) = (0.01) \times 10 = 0.1$

$0.01 \leftarrow P(A \cap B) = \frac{1}{100}$
 $P(A)P(B) = \frac{1}{10} \cdot \frac{1}{10} = \frac{1}{100}$



SON IND.

b) Si A y B SON IND $\rightarrow A$ y $\bar{B}$ SON IND.

$P(A \cap \bar{B}) = P(A)P(\bar{B})$

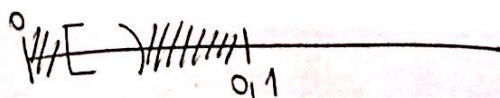
$P(\bar{B}) = 9/10$

$P(A \cap \bar{B}) = \frac{9}{100}$

$P(A)P(\bar{B}) = \frac{1}{10} \cdot \frac{9}{10} = \frac{9}{100} \checkmark$

↓
1ro cero
2do no uno

SON IND.



$0.1 - \frac{1}{100} = \frac{9}{100}$

21)

azar $\rightarrow (0,1)$

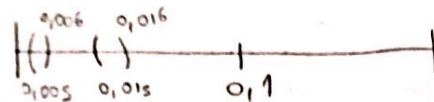
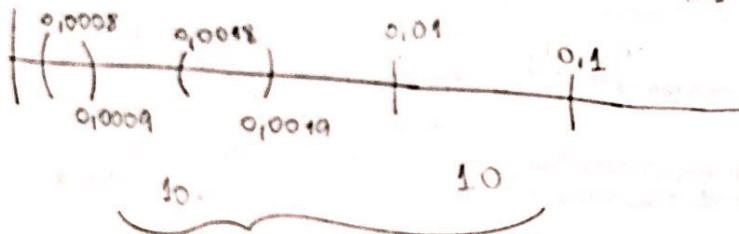
$[0,2, \quad ,0,3)$

- a) "1er dígito es 2" : $A_1 \rightarrow P(A_1) = \frac{1}{10}$
 "2do es 3" : $A_2 \rightarrow P(A_2) = \frac{1}{10}$
 "3ro es 5" : $A_3 \rightarrow P(A_3) = \frac{1}{10}$
 "4to es 8" : $A_4 \rightarrow P(A_4) = \frac{1}{100}$
 (El 1 aparece alguna vez desde el 10 no dígito)

$[0,03, 0,04) \times 10$

$[0,0005, 0,006) \times 100$

0,0008, 0,0009



$$(0,0009 - 0,0008) \times 1000 =$$

$$100 \times 10 = 1000$$

$$P(4^{\text{to}} \text{ es } 8) = \frac{1}{10}$$

$$P(\frac{1}{10}) = 0,0008000001 \rightarrow 1^{\text{er}} \text{ por c/intervalo}$$

$[0,23, 0,24)$

$$P(A_4) = \frac{1}{100}$$

0,235

0,2308

0,035

$$\begin{aligned} P(A_1 \cap A_2) &= P(A_1) P(A_2) = \frac{1}{100} \checkmark \\ P(A_2 \cap A_3) &= P(A_2) P(A_3) = \frac{1}{100} \\ P(A_3 \cap A_4) &= P(A_3) P(A_4) = \frac{1}{1000} \\ P(A_2 \cap A_4) &= P(A_2) P(A_4) = \frac{1}{1000} \\ &\rightarrow 0,0308 \end{aligned}$$

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= \frac{1}{1000} \\ P(A_1 \cap A_2 \cap A_4) &= \frac{1}{104} \\ P(A_2 \cap A_3 \cap A_4) &= \frac{1}{104} \\ P(A_1 \cap A_3 \cap A_4) &= \frac{1}{104} \end{aligned}$$

$$P(A_4 \cap A_1) = P(A_4) P(A_1) = \frac{1}{1000}$$

$$P(A_3 \cap A_1) = P(A_3) P(A_1) = \frac{1}{100}$$

$$P(A_1 \cap A_2 \cap A_3 \cap A_4) = \frac{1}{10^5} \checkmark$$

0,2358000001

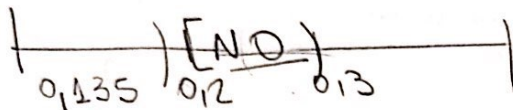
SON IND.

(18)

"el 1ro no es 2" $\rightarrow P(\bar{A}_1) = 1 - P(A_1) = 9/10$.

"el 2do es 3 y 3ro 5" $\rightarrow P(\underbrace{A_2 \cap A_3}_{A_v}) = \frac{1}{100}$

$$P(\bar{A}_1 \cap (A_2 \cap A_3)) = \frac{9}{1000} = P(\bar{A}_1) P(A_2 \cap A_3)$$



son ind.

$(0,136 - 0,135) \rightarrow$ por a. $\rightarrow$ por que el de
 $(0,2) \rightarrow (0,3)$

c) "el 1ro es 1 o 3ro no es 5" $P(\underbrace{A_1 \cup \bar{A}_3}_B) = \frac{1}{10} \cdot \frac{9}{10} = \frac{9}{100}$.

"el 2do es 3 y 4to A_4 " $P(\underbrace{A_2 \cap A_4}_C) = \frac{1}{1000}$

$$P(B \cap C) = \frac{9}{10000}$$

$$0,13 \underline{} 8000001$$

$\times 9$

$$P(\text{no defecto}) = 0,9$$

$$P(\text{defecto}) = 0,01 \text{ o menor}$$

$$P(d) = 0 \rightarrow \text{aceptable}$$

As 300 piezas sin defectos

$$a) P(A) = 1 - \underbrace{0,01}_{\text{prob defecto}}^{300} \approx 1$$

b) 98% aceptable $\rightarrow 0,98 = \text{aceptable}$

$$P(A) = 0,98 = 1 - \underbrace{x^{300}}_{\text{prob defecto}} \rightarrow x \approx 0,987$$

23

2R
1B

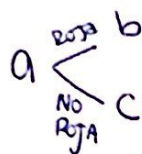
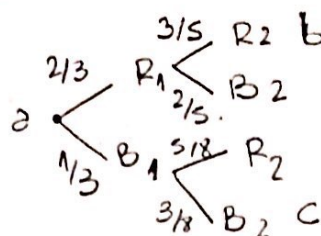
a

3R
2B

b

5R
3B

c

 $R_i, i = 1, 2 \rightarrow i \text{ fue roja}$ $B_i, i = 1, 2 \rightarrow i \text{ fue blanca (moneda)}$

a) $P(B_1) = 1/3$

b) $P(B_2) = \frac{R_1 \cap B_2}{P(B_2)} + \frac{B_1 \cap B_2}{P(B_2)} = \frac{\frac{2}{3} \cdot \frac{2}{5}}{\frac{2}{3} \cdot \frac{2}{5} + \frac{1}{3} \cdot \frac{5}{8}} = \frac{47}{120}$

c) PRIMERA B si 2da ROJA

$$P(B_1 | R_2) = \frac{P(B_1 \cap R_2)}{P(R_2)} = \frac{\frac{1}{3} \cdot \frac{5}{8}}{\frac{2}{3} \cdot \frac{2}{5} + \frac{1}{3} \cdot \frac{5}{8}} = \frac{5/24}{73/120}$$

$$P(B_1 | R_2) = \frac{25}{73}$$

d) alguna sea roja o $R_N \rightarrow \frac{R_1 R_2 | B_1 R_2}{R_1 B_2} \rightarrow \boxed{R_1 \cup R_2}$ otra forma
 $P(R_N) = P(R_1 \cap R_2) \cup P(R_1 \cap B_2) \cup P(B_1 \cap R_2) \rightarrow$ como son
disjuntos la
suma es la
unión.

$$P(R_N) = \frac{2}{3} \cdot \frac{3}{5} + \frac{2}{3} \cdot \frac{2}{5} + \frac{1}{3} \cdot \frac{5}{8} = 7/8$$

e)	<table><tr><td>200R</td></tr><tr><td>100B</td></tr></table>	200R	100B	<table><tr><td>150R</td></tr><tr><td>100B</td></tr></table>	150R	100B	<table><tr><td>125R</td></tr><tr><td>75B</td></tr></table>	125R	75B
200R									
100B									
150R									
100B									
125R									
75B									

antes

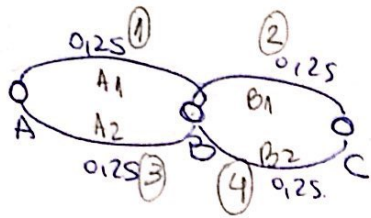
2
1

3
2

5
3

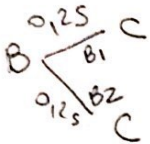
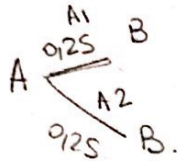
$\rightarrow$ según los mismos
resultados pague
respecto con el total
estén en la misma
proporción.

(2)

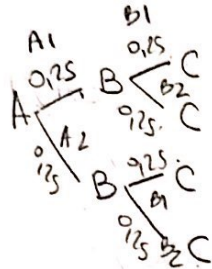


0.25 $\rightarrow$ bloqueado
0.75 $\rightarrow$ ✓

Bloqueado.



B $\rightarrow$ C sabiendo que no hay molestia de A a C.



A = abierto
 $\bar{A}$ = bloqueado.

$$P(BC | \bar{A}) = \frac{P(BC \cap \bar{A})}{P(\bar{A})} = ?$$

caminos de $\bar{A}$

x4 por que va a ser
abierto 3 y 1 abierto
nunca vas a llegar

$$\begin{aligned} &4 (A_1 \bar{B}_1 \bar{B}_2 \bar{A}_2) \\ &2 (\bar{A}_1 \bar{A}_2 B_1 \bar{B}_2) \\ &\bar{A}_1 \bar{A}_2 \bar{B}_1 \bar{B}_2 \end{aligned}$$

$$4 \cdot \left(\frac{3}{4}\right) \left(\frac{1}{4}\right)^3 + 2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^4 = \frac{31}{256}$$

$$P(BC \cap \bar{A}) = P(\bar{A}_1 \cap \bar{A}_2 \cap B_1 \cap B_2) + P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{B}_1 \cap \bar{B}_2) \cdot 2$$

$$= \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right) = 15/256$$

$\bar{A}_1 \bar{A}_2 B_1 B_2$

$\bar{B}_1 \bar{B}_2 A_1 A_2$

$$P(BC | \bar{A}) = 15/31 \approx 0.484$$

PR

$\bar{B}_2 \cap B_1$

1.25

0 y 1

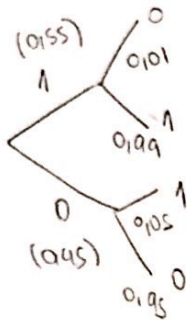
emita

$$P(1) = 0,55$$

$$P(0) = 0,45$$

$$P(0|0) = 0,95$$

$$P(1|1) = 0,99$$



a) $P(1) = 0,55 \cdot 0,99 + 0,45 \cdot 0,05$

$$P(1) = 0,567$$

b) $P(1|1) = \frac{P(1 \cap 1)}{P(1)} = 0,96$

$$P(1 \cap 1) = 0,54$$

1.26

E: emite 1 E: emite 0
R: recebe 1 R: recebe 0

$$P(E) = 0,05$$

$$P(E) = 0,95$$

$$P(R|E) = 0,9 \rightarrow P(R|E) = 0,1$$

c) $P(R|E)? \rightarrow P(E|R) = 0,99$

$$P(E|R) = 0,01$$

$$P(R|E) = \frac{P(R \cap E)}{P(E)}$$

$$P(E|R) = \frac{P(E \cap R)}{P(R)}$$

$$P(R|E) = \frac{P(E \cap R)}{P(E)}$$

$$P(E|R) \cdot P(R) = P(E \cap R) \quad P(E|E) \cdot P(E) = P(E \cap E)$$

III $0,01 \cdot 0,5 \cdot 10^{-3}$ I $P(R) = 0,5 \quad P(E) = 0,05$

II $P(R|E) + P(R|E) = 1$

$$P(R|E) + 5,26 \times 10^{-3} = 1$$

$$P(R|E) = 0,9947$$

$$P(R|E) = \frac{P(R \cap E)}{P(E)}$$

2

equilibrada

2C
①

1C
1S
②

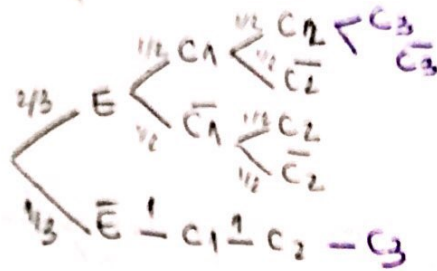
1C
1S
③

S = 2C
C = 1C1A

E = equilibrada

a)

$$P(C_2|C_1) = \frac{P(C_1 \cap C_2)}{P(C_1)}$$



$$P(C_1) = \frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1 = \frac{2}{3}$$

$$P(C_1 \cap C_2) = \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1 \cdot 1 = 0,5$$

$$P(C_2|C_1) = \frac{3}{4}$$

b)

$$P(C_1) = \frac{2}{3}$$

$$P(E|C_1) = \frac{P(E \cap C_1)}{P(C_1)} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

$$P(E|C_1) = \frac{1}{2}$$

$$P(\bar{E}|C_1) = \frac{1}{2}$$

$$c) \quad P(\bar{E} | C_1 \cap C_2) = \frac{P(\bar{E} \cap (C_1 \cap C_2))}{P(C_1 \cap C_2)} = \frac{\frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{2}}{\frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{3}}$$

$$\left[P(\bar{E} | C_1 \cap C_2) = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3} \right]$$

$$d) \quad \left[P(\bar{E} | C_1 \cap C_2 \cap C_3) = \frac{\frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}}{\frac{1}{12} + \frac{1}{3}} = \frac{1}{5} \right]$$

(23)

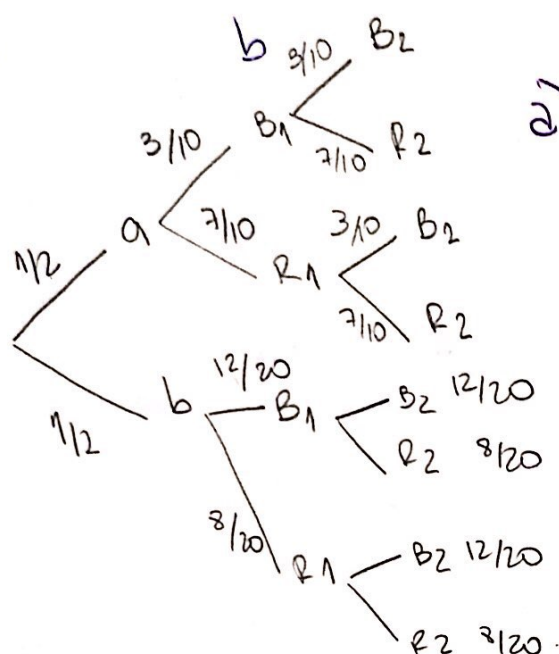
28

2 con rep 0

3B
7R

12B
8R

a



$$2) P(B_2|B_1) = \frac{P(B_2 \cap B_1)}{P(B_1)}$$

$$P(B_1) = \frac{1}{2} \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{12}{20} = \frac{9}{20}$$

$$P(B_2 \cap B_1) = \frac{1}{2} \cdot \frac{3}{10} \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{12}{20} \cdot \frac{12}{20} = \frac{9}{40}$$

$$P(B_2|B_1) = \frac{1}{2}$$

b) "primera blanca", "segunda blanca"

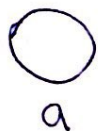
$$P(B_1 \cap B_2) = P(B_1) P(B_2)$$

$$\frac{9}{40} = \frac{9}{20} \cdot \frac{9}{20}$$

$\frac{9}{40} \neq \frac{81}{400}$ — NO SON IND.

$$P(B_2) = \frac{1}{2} \cdot \frac{3}{10} \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{7}{10} \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{12}{20} \cdot \frac{12}{20} + \frac{1}{2} \cdot \frac{8}{20} \cdot \frac{12}{20} = \frac{9}{200} + \frac{21}{200} + \frac{9}{50} + \frac{3}{25} = \frac{9}{20}$$

(24)



a

$$P(C) = 1/2$$

$$P(\bar{C}) = 1/2$$



b

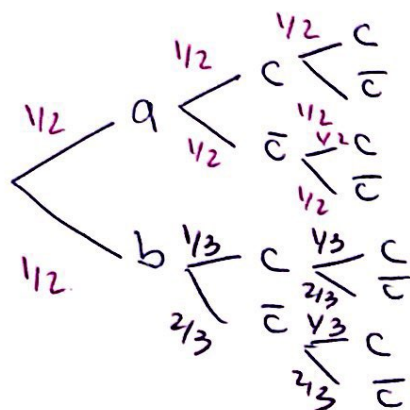
$$P(C) = 1/3$$

$$P(\bar{C}) = 2/3$$

A_1 = seleccionar el 1er nro

A_2 = " " 2do nro.

B = se elige la moneda a



Dado.
a1) B, A_1 y A_2 son ind?

a2) A_1 y A_2 son ind?

a1)
$$P(A_1 \cap A_2 | B) = \frac{P(A_1 \cap A_2)}{P(B)} = \frac{1/8}{1/2} = 1/4$$

$$P(B) = 1/2$$

$$P(A_1 | B) = \frac{P(A_1 \cap B)}{P(B)} = 1/4 \div 1/2$$

$$P(A_1 \cap A_2) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

$$P(A_2 | B) = \frac{\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}}{1/2} = 1/2$$

$$P(A_2 \cap A_1 | B) = P(A_1 | B) \cdot P(A_2 | B)$$

$$1/4 = 1/2 \cdot 1/2 \checkmark \text{ son ind } A_1 \text{ y } A_2.$$

a2)

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$$

$$P(A_1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{3} = \frac{5}{12}$$

$$\frac{13}{72} \neq \frac{25}{144} \text{ NO SON IND.}$$

$$P(A_2) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} +$$

$$P(A_1 \cap A_2) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{13}{72}$$

$$\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{5}{12} \text{ (25)}$$

A_1 = sale caro 1^{ro}
 A_2 = sale 2^{do}.

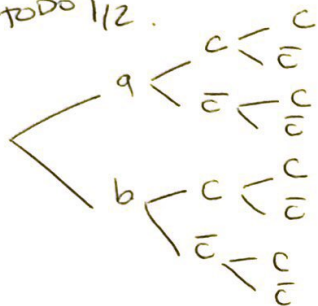
B = sale 1 ceca.

moneda
 equilibrada.

¿ A_1 y A_2 son ind?

¿Dado B , A_1 y A_2 son?

todo $1/2$.



$$P(A_1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$$P(A_2) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

$$P(A_1 \cap A_2) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

son ind ✓

$$P(A_1 \cap A_2 | B) = \underbrace{P(A_1 | B) P(A_2 | B)}_{\neq 0}$$

no son ind ✓